LAB #02

MELISA GUZMAN MUÑOZ

DISEÑO DE DATOS Y ALGORITMOS

ESCUELA COLOMBIANA DE INGENIERIA

BOGOTÁ, 2026

EJERCICIOS

1. Ordene de menor a mayor

$$n! > e^n > 2^n > 2^{n-1} > n - n^3 + 7n^5 > n^3 > n^2 + logn > n^2 > n^{1+\varepsilon} > n\,log\,n > n > \sqrt{n} > (log\,n\,)\,\hat{}\,2 > ln\,n > log\,n > log\,log\,n$$

2. Determinar el resultado como una función de n y representar el peor caso de ejecución con notación Big oh:

FUNCION	COSTO	TIEMPO
r := 0	C1	1
for i := 1 to n-1 do	C2	n
For j := i + 1 to n do	C3	n(n-1)
for k:= 1 to j do	C4	(n-1)(n-1)($\sum_{k=1}^{j} tk$)
r:= r+1	C5	(n-1)(n-1)($\sum_{k=1}^{j} tk - 1$)
Return(r)	C6	1

$$T(n) = c1 + c2 * n + c3 * \mathrm{n(n-1)} + c4 * \text{(n-1)(n-1)}(\textstyle\sum_{k}^{j} tk) + C5 * \mathrm{(n-1)(n-1)}(\textstyle\sum_{k}^{j} tk - 1) + c6 * 1$$

Desarrollando la escuacioón obtenemos: $T(n) = c1 + c2 * n + c3 * n * (n-1) + c4 * (n-1)^2 * \sum_{k=1}^{j} t_k + c_5 * (n-1)^2 * \sum_{k=1}^{j} (t_k - 1) + 1$

Las sumatorias se desarrollan así para el mejor caso (t=1) : $\sum_{k=1}^{j} t_k = \sum_{k=1}^{j} 1 = j$

$$\sum_{k=1}^{j} (t_k - 1) = \sum_{k=1}^{j} 0 = 0$$

$$T(n) = c_1 + c_2 * n + c_3 * n * (n-1) + c_4 * (n-1)^2 * j + 1$$

$$T_m(n) = \theta(n^3)$$

Para el peor caso:

$$\Sigma_{(k=1)}^{(n-1)} t_k = \Sigma_{(k=1)}^{(n-1)} k = n(n-1)/2$$

$$\Sigma_{(k=1)}^{(n-1)} (t_k - 1) = \Sigma_{(k=1)}^{(n-1)} (k-1) = n(n-1)/2 - (n-1) = (n-1)(n-2)/2$$

$$T_p(n) = c_1 + c_2 n + c_3 n(n-1) + c_4 (n-1)^2 (n(n-1)/2) + c_5 (n-1)^2 \big((n-1)(n-2)/2\big) + c_6$$

$$T_p(n) = O(n^4)$$

Para tk = 1 (mejor caso): $T(n) = \theta(n^3)$

Para tk (peor caso): $T(n) = \theta(n^4)$

FUNCION	COSTO	TIEMPO
r := 0	C1	1
for i := 1 to n do	C2	n
For j := 1 to i do	C3	(n-1)$\left(\sum_{j=1}^{i} tj\right)$
for k:= j to i+j do	C4	(n-1) $(\sum_{j=1}^{i} tj - 1)(\sum_{k=j}^{i+j} tk)$
r:= r+1	C5	(n-1)$(\sum_{j}^{i} tj - 1)(\sum_{k=j}^{i+j} tk - 1)$
Return(r)	C6	1

$T(n) = c1 + c2 * n + c3 *$(n-1)$\left(\sum_{j=1}^{i} tj\right) + c4 *$(n-1) $(\sum_{j=1}^{i} tj - 1)(\sum_{k=j}^{i+j} tk)$ + c5 * (n-1)$(\sum_{j}^{i} tj - 1)($ $\sum_{k=j}^{i+j} tk - 1) + c6$

Mejor caso (tj=1 y tk=1)

$$\sum_{j=1}^{i} t_j = \sum_{j=1}^{i} 1 = i$$

$$\sum_{k=j}^{i+j} t_k = \sum_{k=j}^{i+j} 1 = (i + j - j + 1) = i + 1$$

$$T_{\text{mejor}}(n) = c_1 + c_2 n + c_3 (n-1) i + c_4 (n-1)(i-1)(i+1) + c_5 (n-1)(i-1) i + c_6$$

Suponinedo que i = n -1 (como em insertion sort)

$$T_{\text{mejor}}(n) = c_1 + c_2 n + c_3 (n-1)^2 + c_4 (n-1)(n-2) n + c_5 (n-1)^2 (n-2) + c_6$$

$$T_{\text{mejor}}(n) = O(n^3)$$

Peor caso ($t_j = j$y $t_k = k$)

$$\sum_{j=1}^{i} t_j = \sum_{j=1}^{i} j = \frac{i(i+1)}{2}$$

$$\sum_{k=j}^{i+j} t_k = \sum_{k=j}^{i+j} k = \frac{(i+j)(i+j+1)-(j-1)j}{2}$$

$$\sum_{k=j}^{i+j} k = \Theta(i^2)$$

$$\left(\sum_{j=1}^{i} t_j - 1\right) = \frac{i(i+1)}{2} - 1 = \Theta(i^2)$$

$$c_5(n-1)\left(\sum_{j=1}^{i} t_j - 1\right)\left(\sum_{k=j}^{i+j} t_k - 1\right) = (n-1)\Theta(i^2)\Theta(i^2) = \Theta\big((n-1)i^4\big)$$

Si i = n-1

$$T_{\text{peor}}(n) = O(n^5)$$

FUNCION	COSTO	TIEMPO
r := 0	C1	1
for i := 1 to n do	C2	n
For j := 1 to i do	C3	(n-1)$\left(\sum_{j=1}^{i} tj\right)$
for k:= j to i+j do	C4	(n-1) $(\sum_{j=1}^{i} tj - 1)(\sum_{k=j}^{i+j} tk)$
For l := 1 to i+j – k do	C5	(n-1)$(\sum_{j}^{i} tj - 1)(\sum_{k=j}^{i+j} tk - 1)(\sum_{l=1}^{i+j-k} tl)$
r:= r+1	C6	(n-1)$(\sum_{j}^{i} tj - 1)(\sum_{k=j}^{i+j} tk - 1)(\sum_{l=1}^{i+j-k} tl - 1)$
Return(r)	C7	1

$$T(n) = c1 + c2 * n + c3 * (n-1)\,(\sum_{j=1}^{i} tj) + c4 * (n-1)\,(\sum_{j=1}^{i} tj - 1)(\sum_{k=j}^{i+j} tk + c5 * (n-1)(\sum_{j}^{i} tj - 1)(\sum_{k=j}^{i+j} tk - 1)(\sum_{l=1}^{i+j-k} tl) + c6 * (n-1)(\sum_{j}^{i} tj - 1)(\sum_{k=j}^{i+j} tk - 1)(\sum_{l=1}^{i+j-k} tl - 1) + c7$$

$$T(n) = c_1 + c_2 n + c_3(n-1)\left(\sum_{j=1}^{i} t_j\right) + c_4(n-1)\left(\sum_{j=1}^{i} t_j - 1\right)\left(\sum_{k=j}^{i+j} t_k\right)$$
$$+ c_5(n-1)\left(\sum_{j=1}^{i} t_j - 1\right)\left(\sum_{k=j}^{i+j}(t_k - 1)\left(\sum_{l=1}^{i+j-k} t_l\right)\right)$$
$$+ c_6(n-1)\left(\sum_{j=1}^{i} t_j - 1\right)\left(\sum_{k=j}^{i+j}(t_k - 1)\left(\sum_{l=1}^{i+j-k} (t_l - 1)\right)\right) + c_7$$

Mejor caso:

$$t_j = 1,$$

$$t_k = 1,$$

$$t_l = 1$$

$$\sum_{j=1}^{i} t_j = \sum_{j=1}^{i} 1 = i$$

$$\sum_{k=j}^{i+j} t_k = \sum_{k=j}^{i+j} 1 = i + 1$$

$$\sum_{k=j}^{i+j}(t_k - 1) = \sum_{k=j}^{i+j}(1 - 1) = 0$$

$$T_{\text{mejor}}(n) = c_1 + c_2 n + c_3(n-1)i + c_4(n-1)(i-1)(i+1) + c_7$$

$$T_{\text{mejor}}(n) = O(n^3)$$

Peor caso:

$$\sum_{j=1}^{i} t_j = \sum_{j=1}^{i} j = \frac{i(i+1)}{2} = \Theta(i^2)$$

$$\sum_{k=j}^{i+j} t_k = \sum_{k=j}^{i+j} k = \frac{(i+j)(i+j+1) - (j-1)j}{2} = \Theta(i^2)$$

$$S_3 = \sum_{k=j}^{i+j}(t_k - 1)\left(\sum_{l=1}^{i+j-k} t_l\right) = \sum_{k=j}^{m}(k-1)\left(\sum_{l=1}^{m-k} l\right)$$

$$S_4 = \sum_{k=j}^{i+j} (t_k - 1) \left(\sum_{l=1}^{i+j-k} (t_l - 1) \right) = \Theta(i^4)$$

$$\left(\sum_{j=1}^{i} t_j - 1 \right) = \Theta(i^2)$$

$$T_{\text{peor}}(n) = \Theta\big((n-1)(n-1)^6\big) = \Theta(n^7)$$

$$T_{\text{peor}}(n) = O(n^7)$$

3. Implementar el algoritmo de insertion sort para ordenar en orden descendente en vez de ascendente.

```python
def insertion_sort_desc(A):
    for j in range(1, len(A)):
        key = A[j]
        i = j - 1
        while i >= 0 and A[i] < key:
            A[i + 1] = A[i]
            i -= 1
        A[i + 1] = key
    return A
```